

AUTOMATIC DOORBELL

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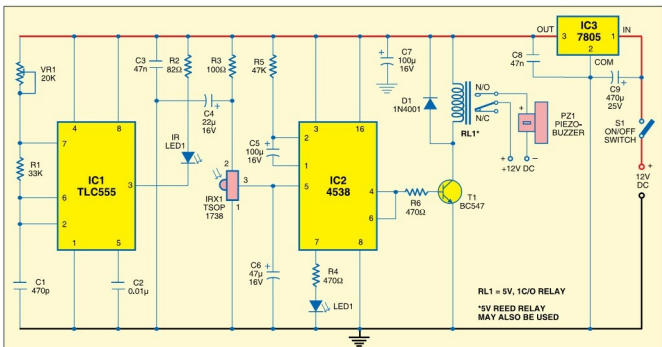
This doorbell rings automatically when someone enters the premises of your house. It comprises an infrared LED transmitter unit, a receiver unit and a piezobuzzer unit. The IR transmitter and receiver units are installed on the opposite pillars of the entrance such that the IR rays transmitted from the IR LED fall directly on

from a suitable 12V compact battery. The 12V DC input power supply is regulated by three-pin, fixed-voltage regulator IC 7805 (IC3). Capacitors C7, C8 and C9 allow for perfect buffering and noise suppression. Switch S1 is used for power-'on'/'off'. LED1 indicates that the automatic doorbell is in standby mode.

The transmitter is built around IC TLC555 (IC1), which is configured

are intercepted and the high-to-low pulse produced at pin 5 of IC2 triggers the monostable (IC2). Consequently, output pin 6 of IC2 goes high and the relay (RL1) energises for a short duration of around eight seconds (which is determined by the in-circuit values of resistor R5 and capacitor C5).

Transistor T1 drives the relay and free-wheeling diode D1 protects it from



the receiver. The piezobuzzer is to be installed inside your room. When someone crosses the entrance, the IR rays falling on the receiver are intercepted and the piezobuzzer sounds.

The circuit is very simple and consists of a power supply built around 5V regulator 7805, an infrared transmitter built around TLC555 and IR LED, an infrared receiver TSOP1738, a monostable built around IC 4538 with controlled relay-driver transistor T1, relay RL1 and piezobuzzer PZ1.

The 12V power supply for the circuit is derived from the AC mains by using a small 12V DC (750mA) adaptor. Else, you can power the circuit

as an astable multivibrator with an operating frequency of approximately 38 kHz. Components VR1, R1 and IC1 determine this frequency. Output pulses from pin 3 of IC1 drive IR LED1. Resistor R2 limits the operating current through IR LED1.

Normally, when the infrared rays from IR LED1 are falling on IRX1 receiver module (TSOP1738), output pin 3 of IRX1 remains high and output pin 6 of IC2 remains low. As a result, relay RL1 remains de-energised and the piezobuzzer does not produce any sound.

When someone enters through the gate of your house, the infrared rays falling on the receiver module

the back emf generated at the time of relay switching. The piezobuzzer is connected to the normally-open contacts of relay RL1. When the relay energises, standby LED1 goes off and the piezobuzzer sounds.

After construction, enclose the circuit in a suitable cabinet. Fix IR LED1 and IR receiver module IRX1 on the opposite pillars of the gate such that the IR rays from IR LED1 fall directly on IR receiver module (TSOP 1738). Fit the IR LED with a proper hood to ensure that the rays do not spill out sideways. Fix the standby LED (LED1) on the front panel of the cabinet. It indicates that the automatic doorbell is

